

DropBox

functional requirements:

- user should be able to upload/download any file based on his preference,
- user should be able to search for any file among the existing files based on his preference,
- user should be able to upload multiple files at a time simultaneously

non-functional requirements:

- data-partitioning,.
- load balancing,
- fault tolerance.
- vertical scalability,
- horizontal scalability.
- low latency,
- high throughput,
- CAP theorem(Here eventual consistency is fine , but here we need 99.99% uptime i.e Availability>>Consistency)

Core Entities:

- user,
- file,
- history,

API:

POST:/upload

```
{  
  userID,  
  filemetadata  
}
```

GET:/search/{fileID}

HLD :

